

AWI-ESM3 (v3.4, CORE3 / TCO95) PI-control tuning campaign

Tuning_test_06* — progress report

Evaluation via release_evaluation_tool2 (part2 / part8 / part4)

2026-06-30 (rev. 2026-07-06: restructured into three parts; decisive feedback-free re-spin shows the boreal deficit is LPJG tree-vs-grass *competition*, not climate)

Part I

Tuning away the high TOA imbalance

*We set out to cut the AWI-ESM3 PI-control's $+1.5 \text{ W/m}^2$ TOA imbalance. Every effective lever does it by **cooling** the model, and the aggressive coupling/ocean-mixing levers cut the imbalance hardest — but push the planet too cold against ERA5. This part documents that trade-off across 22 runs and traces the aggressive lever to a Southern-Ocean ocean-convection mechanism.*

1 Goal & setup

Reduce the net top-of-atmosphere (TOA) radiative imbalance of the AWI-ESM3 pre-industrial control so it equilibrates faster, *without* degrading the CMIP6-normalised performance index (CMPI, **part4**) or inflating the 2 m-temperature RMSE vs ERA5 (**part8**). 22 experiments, each a 30-year segment (model years 1350–1379) of the CORE3/TCO95 configuration (FESOM2 CORE3 mesh, OpenIFS TCO95/A096, LPJ-GUESS), all branched from one **Baseline**. The baseline PI-control still drifts with a positive TOA imbalance of $\approx +1.5 \text{ W/m}^2$ (energy *into* the system — the signature of unequilibrated spin-up and/or ongoing ocean heat uptake). **A caution on the target.** The tuning levers change the model's *equilibrium*, not its rate of approach, so “lower imbalance” is a useful *ranking* objective but not automatically “more equilibrated”: a low-imbalance-but-cold state is not obviously better than a high-imbalance-but-unbiased one. This is why the campaign's real aim (§8) is a *warm-floor* state where the imbalance is cut by a *moderate* lever, not by cooling into a cold bias — i.e. we treat imbalance and 2 m-T bias as joint targets, not imbalance alone.

Structural note. Almost every non-baseline run also carries a *common melt-pond shift* (albpnd $0.2 \rightarrow 0.28$, rfracmax $1.0 \rightarrow 0.75$, pndaspect $0.8 \rightarrow 1.3$). Differences below are each run's *distinguishing* change on top of that.

Three parts. **(I)** evaluates the 22 imbalance-tuning runs and exposes the imbalance $\leftrightarrow$ cold-bias trade-off, tracing the aggressive lever to a Southern-Ocean mechanism. **(II)** shows that the cold bias which makes that lever costly is largely *not* the tuning but a **structural** LPJ-GUESS boreal-forest collapse (§6) — a *counter-lever*: fixing it warms the NH high latitudes, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. A decisive feedback-free experiment pins that fix to vegetation, not climate. **(III)** combines them into a work plan: raise the structural warm floor, then cut the imbalance (§8).

2 Headline: the imbalance $\leftrightarrow$ cold-bias trade-off

Every lever that meaningfully lowers the TOA imbalance does so by *cooling* the model, which worsens the cold bias against ERA5. Two families separate cleanly: the sea-ice/albedo/melt-pond changes move gently down-left (small imbalance gain, small bias cost); the coupling + ocean-mixing changes cut the imbalance hard (30–57%) but push the planet 1.7–2.1 K too cold.

The cold-bias axis is two things at once. It mixes (i) a *shared structural floor* — every run, including the baseline and the “clean” melt-pond runs, carries a strong Siberia/N. Canada cold bias that is *not* a tuning artefact but an LPJ-GUESS boreal-forest-cover bug (§6); and (ii) the *differential*, tuning-induced global cooling that the aggressive runs add on top. The structural floor is the same for all points and is independently fixable; the tuning only moves runs vertically relative to it. This is why the “warming lever” in §8 is the LPJG fix, not another atmospheric retune.

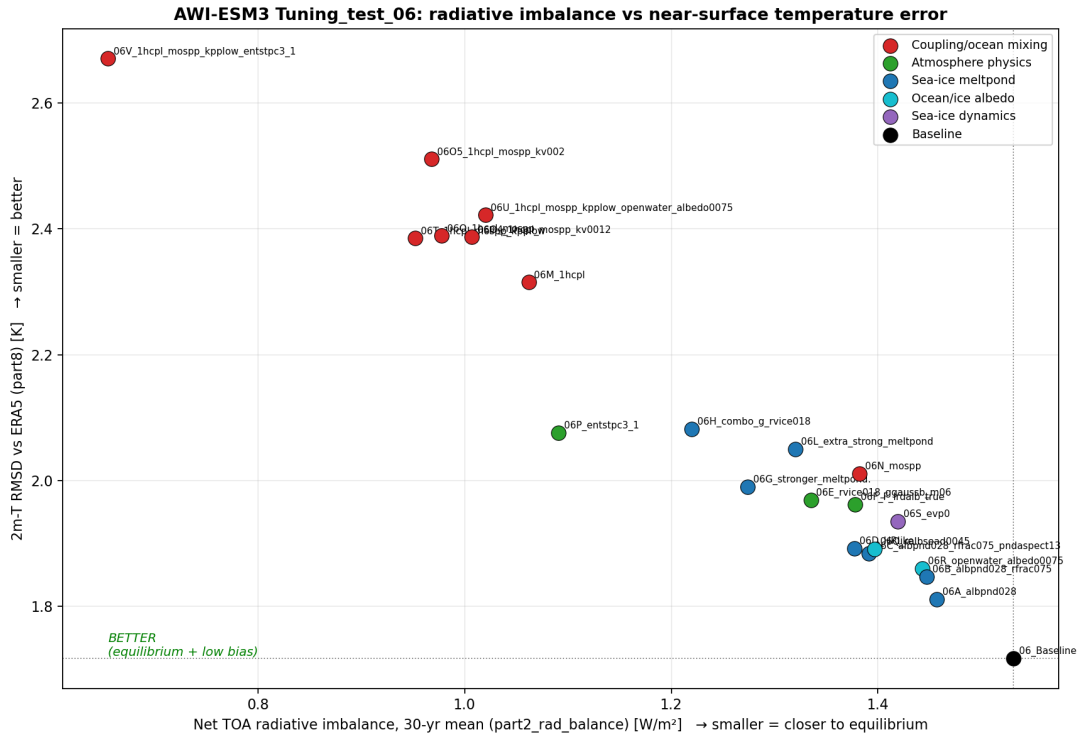


Figure 1: Net TOA imbalance (30-yr mean, **part2**) vs 2m-T RMSD against ERA5 (**part8**). Dotted lines = baseline; lower-left is better. Colour = parameter family.

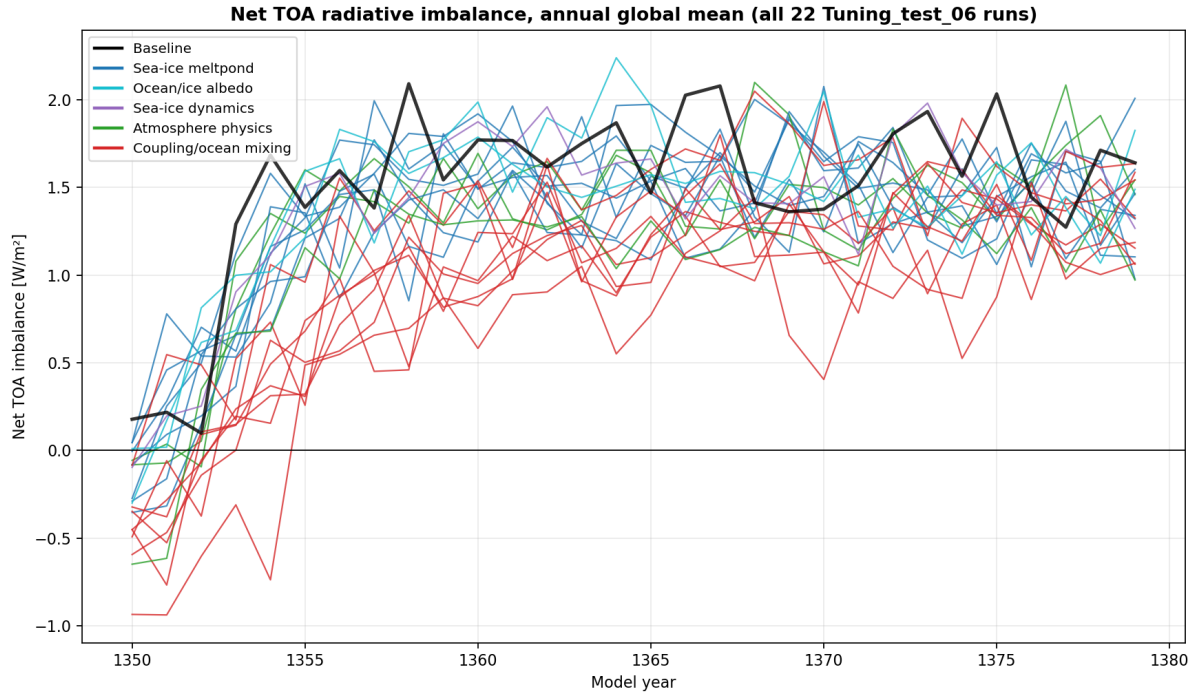


Figure 2: Annual global-mean net TOA imbalance, all 22 runs (independent CDO calculation; matches **part2**). None has flattened within 30yr, so treat the mean level as a *ranking*, not a final equilibrium value.

3 Master metric table (all 22 runs, sorted by TOA imbalance)

Run	TOA imb. [W/m ²]	vs base	2m RMSD [K]	2m bias [K]	CMPI
06V_1hcpl_mospp_kpplow_entstpc3_1	0.655	−57%	2.671	−2.124	0.803
06T_1hcpl_mospp_kpplow	0.952	−38%	2.385	−1.791	0.773
06O5_1hcpl_mospp_kv002	0.968	−37%	2.511	−1.904	0.788
06O_1hcpl_mospp	0.978	−36%	2.389	−1.774	0.772
06O4_1hcpl_mospp_kv0012	1.007	−34%	2.387	−1.807	0.781
06U_1hcpl_mospp_kpplow_openw0075	1.020	−33%	2.422	−1.845	0.776
06M_1hcpl	1.062	−31%	2.315	−1.658	0.773
06P_entstpc3_1	1.091	−29%	2.076	−1.358	0.759
06H_combo_g_rvice018	1.220	−20%	2.082	−1.394	0.757
06G_stronger_meltpond	1.274	−17%	1.990	−1.274	0.753
06L_extra_strong_meltpond	1.320	−14%	2.050	−1.332	0.751
06E_rvice018_ggaussb_m06	1.335	−13%	1.969	−1.231	0.754
06D_HRlike	1.377	−10%	1.892	−1.146	0.750
06F_lrdalb_true	1.378	−10%	1.962	−1.219	0.752
06N_mospp	1.382	−10%	2.011	−1.310	0.751
06C_albpnd028_rfrac075_pndaspect13	1.391	−9%	1.884	−1.109	0.751
06Q_ralbsead0045	1.397	−9%	1.891	−1.154	0.748
06S_evp0	1.419	−7%	1.935	−1.191	0.745
06R_openwater_albedo0075	1.443	−6%	1.860	−1.127	0.744
06B_albpnd028_rfrac075	1.447	−5%	1.847	−1.062	0.741
06A_albpnd028	1.457	−5%	1.811	−1.004	0.748
06_Baseline	1.531	—	1.717	−0.939	0.745

TOA = 30-yr global-mean net TOA from `part2_rad_balance.py`. RMSD/bias from `part8` methodology (cos-lat weighted, model–ERA5, yrs 1354–1379); cross-checked vs the tool’s on-figure values (baseline 1.717 vs 1.721; 06V 2.671 vs 2.678). CMPI from `part4_cmpi.py`: < 1 = better than CMIP6 ensemble-mean error; lower is better.

The per-run distinguishing parameter change (which knob each 06* run alters on top of the common melt-pond shift) is catalogued in App. A.

4 Core diagnostics: radiation, 2 m temperature, CMPI

4.1 Radiation budget (part2)

Surface net flux tracks TOA to within $\approx 0.1 \text{ W/m}^2$ in every run — i.e. the atmospheric column stores essentially no energy on these multi-year means, as expected. Note this does *not* by itself localise the *cause*: a TOA $\approx$ surface match is equally consistent with a genuine TOA radiative error and with ocean heat uptake / unequilibrated spin-up drift passing cleanly through the surface (§5, §6.3). What it rules out is an atmosphere-only bookkeeping leak, so the imbalance and the surface energy budget can be read interchangeably. Baseline $\approx +1.53 \text{ W/m}^2$; the coupling+mixing stack drives it down to $+0.66 \text{ W/m}^2$ (06V).

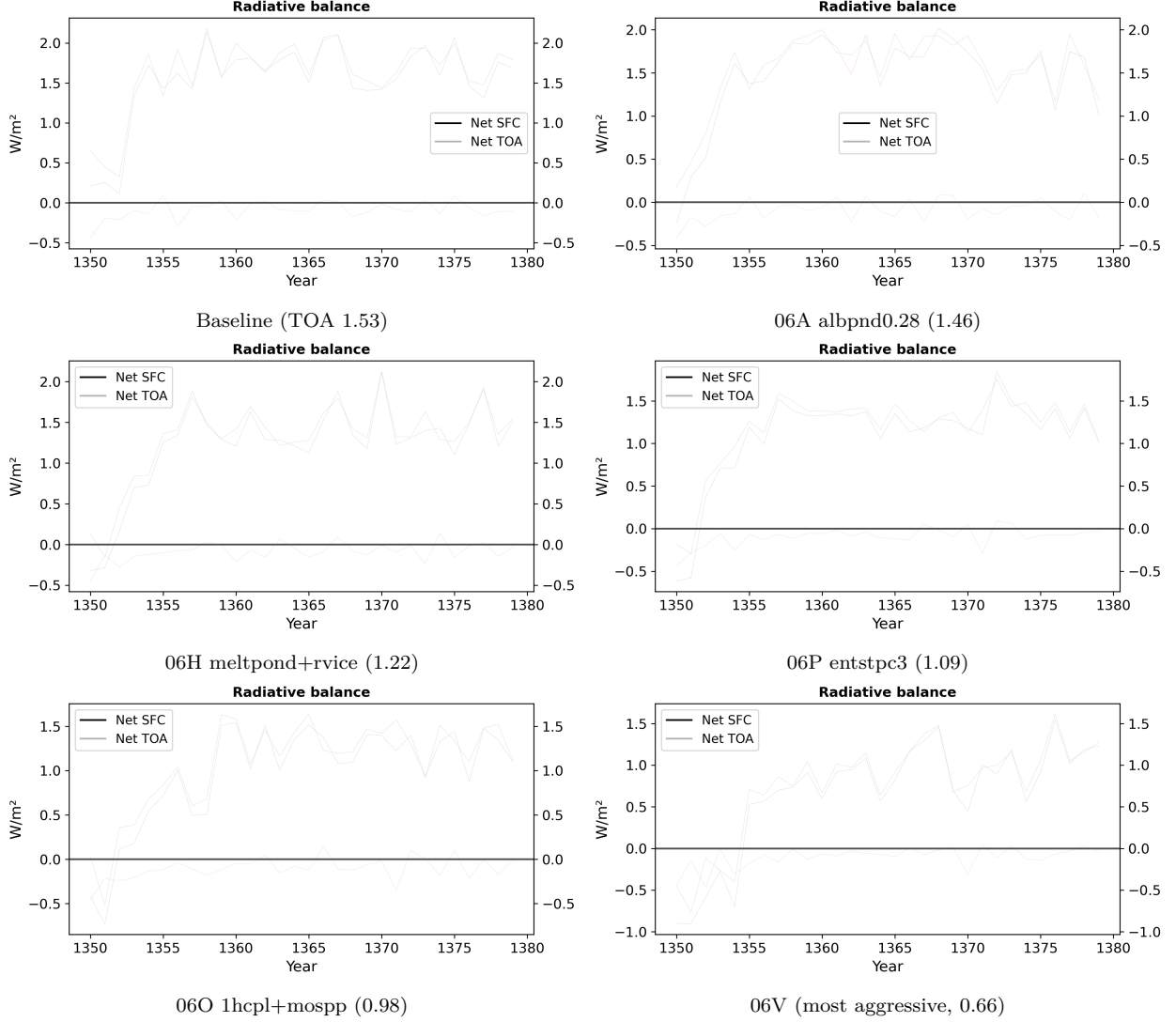


Figure 3: part2 radiation budget: grey = net TOA, blue = net surface, dark = their difference (W/m^2).

4.2 2 m temperature bias vs ERA5 (part8)

Baseline: cold over NH continents, warm over the Southern Ocean and tropical upwelling. The aggressive coupling runs remove the Southern-Ocean warm bias but turn the whole planet cold (06V is cold almost everywhere). *Caveat:* this is PI-control vs present-day ERA5, so $\approx 0.5\text{--}0.7\text{ K}$ of global cold offset is physically expected; the coupling runs' -1.7 to -2.1 K is clearly too much, the melt-pond family (-1.0 to -1.4 K) is more defensible.

The deep **Siberia / N. Canada cold lobe** is present in *every* panel — baseline and the clean melt-pond runs included — and barely changes between them: it is the structural LPJ-GUESS forest-cover signal (§6), not a product of the imbalance tuning. The aggressive coupling runs add a broad, near-global cooling *on top* of that fixed lobe.

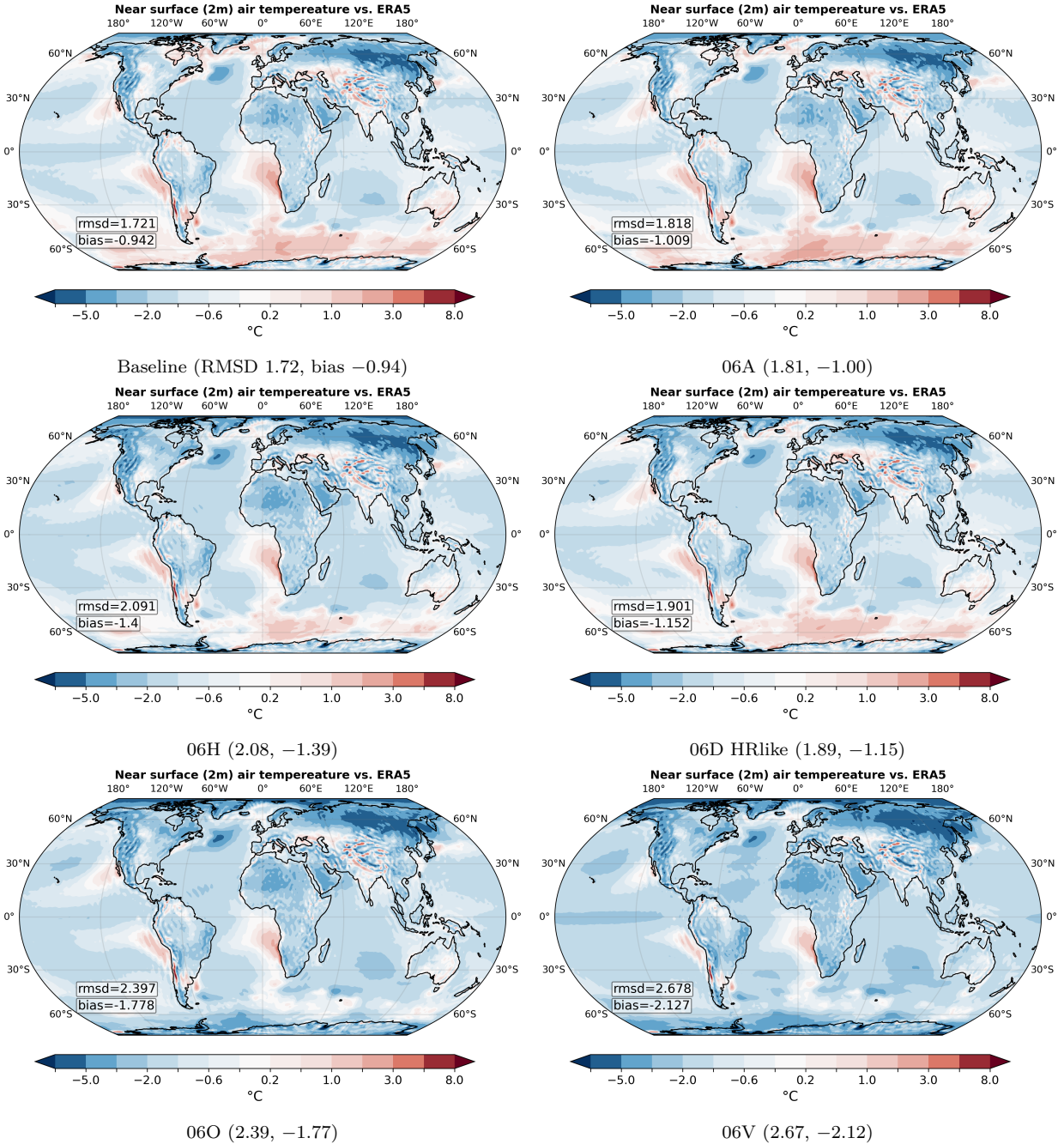


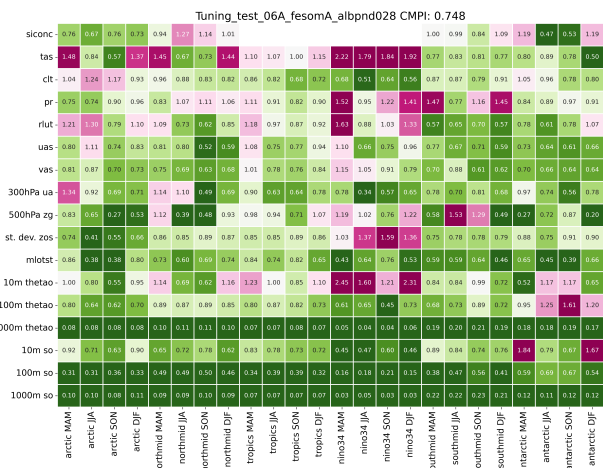
Figure 4: part8 2m air-temperature bias (model – ERA5), EqualEarth, $\pm 8\text{ K}$ scale.

4.3 CMPI (part4)

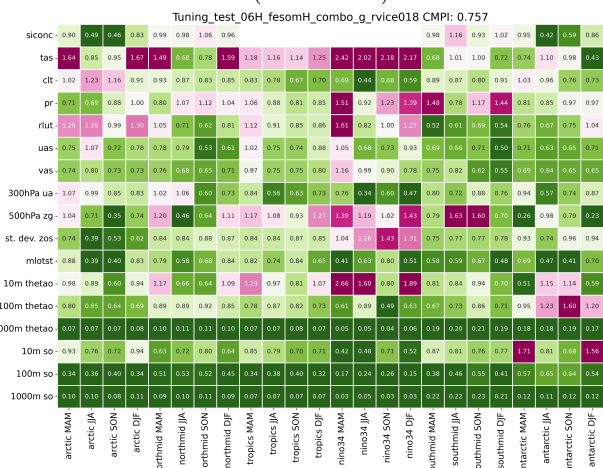
CMPI integrates 13 fields (siconc, tas, clt, pr, rlut, uas, vas, ua@300, zg@500, zos, mlotst, thetad, so) over regions/seasons, normalised against 30 CMIP6 models; < 1 = **better than the CMIP6 ensemble-mean error**. Run on a subset spanning the trade-off front (now complete for all 22 runs).

How to read this (important). CMPI here compares a *PI-control* run to *present-day* observations, so a moderate global cold offset — and hence some CMPI rise — is *expected* and is not by itself a disqualifier. The informative question is not “did CMPI go up?” but “*where* did the error change, and is that consistent with the parameter that was altered?” A sea-ice-albedo change should move **siconc** in the polar regions and little else; if it instead degrades the tropics, that is a red flag. The per-variable / per-region breakdown below is the tool for that judgement; the global number is a summary, not a verdict.

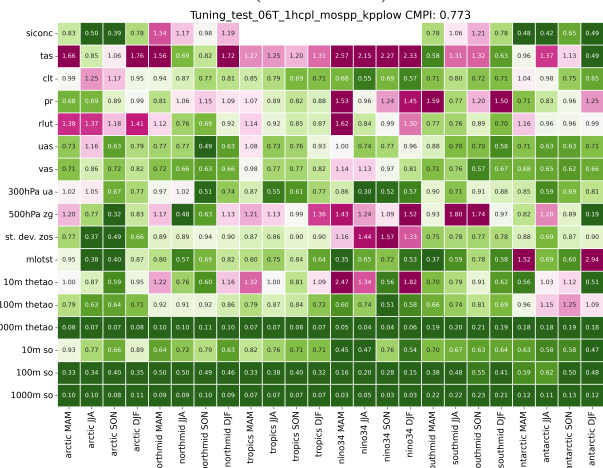
Run	CMPI	TOA imb.	2m RMSD
06_Baseline	0.745	1.531	1.717
06A_albpnd028	0.748	1.457	1.811
06D_HRlike	0.750	1.377	1.892
06H_combo_g_rvice018	0.757	1.220	2.082
06O_1hcpl_mospp	0.772	0.978	2.389
06T_1hcpl_mospp_kpplow	0.773	0.952	2.385
06V_...entstpc3_1	0.803	0.655	2.671



06A (CMPI 0.748)



06H (CMPI 0.757)



06T (CMPI 0.773)

7

4.3.1 Where does the error change? (per-variable / per-region)

The maps below show each run’s CMPI fractional error *minus baseline* (red = worse, blue = better). This is where the PI-vs-PD caveat earns its keep: the global CMPI rises monotonically ($0.745 \rightarrow 0.803$), but the *pattern* of change tells a much more favourable story for the targeted tunings.

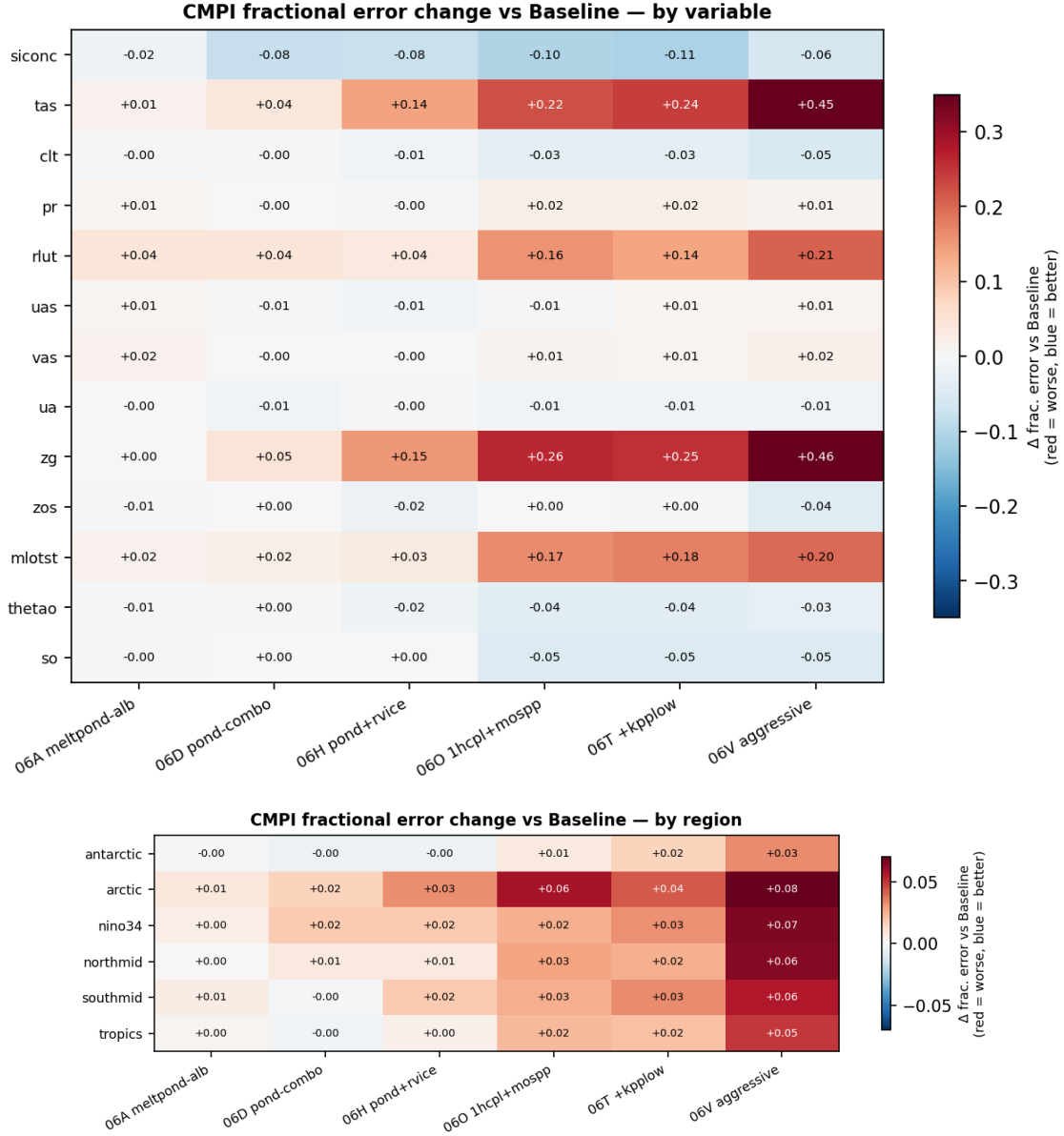


Figure 6: Change in CMPI fractional error vs Baseline, by variable (top) and region (bottom).

Reading the breakdown:

- **06A / 06D (melt-pond):** improve **siconc** ($-0.02/-0.08$) with essentially *zero* collateral elsewhere and a flat region map — a cleanly targeted change. These should be regarded as “no degradation where it matters,” despite the tiny global CMPI uptick.
- **06O / 06T (mospp + kpplow):** genuinely *improve* the ocean interior (**thetao**, **so**: $\approx -0.04/-0.05$), **siconc** and **clt** — exactly where ocean-mixing should act. But they *degrade* **tas/zg/rlut** ($+0.16$ to $+0.26$) and even their own target **m1otst** ($+0.17$), and the region map reddens *everywhere incl. tropics and Niño-3.4* — i.e. the cooling leaks well beyond where an ocean-mixing parameter has any business acting. *That* is the warning sign, more than the global number.
- **06V (+entstpc3):** largest, most widespread atmospheric degradation (**tas** $+0.45$, **zg** $+0.46$); the deep-convection change destabilises the whole atmospheric column.

5 Southern-Ocean mechanism (adapted from a270234’s notebooks)

To see *why* the ocean-mixing/coupling levers act as they do, two of a270234’s Weddell-Sea diagnostics (mld_sic_sit_compare, temp_fesom_core3_WWS) were re-pointed at the coupled Tuning_test_06 runs (same CORE3 mesh). Panels show the Baseline absolute field plus each run *minus* Baseline, austral winter (September), yr1370–79. (The temp_fesom_core3_WWS tripyview transect could not decode these runs’ pre-1582 gregorian calendar, so an equivalent pyfesom2 depth–latitude section was used.)

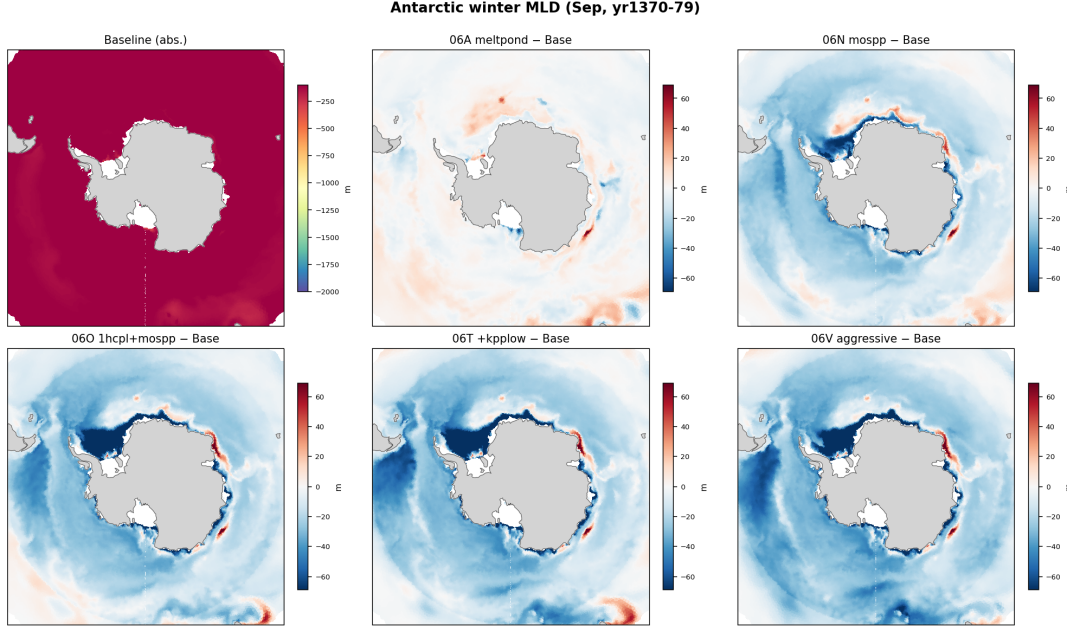


Figure 7: Antarctic September mixed-layer depth (MLD3): Baseline *absolute* (top-left, own scale), others = *anomaly* vs Baseline on a symmetric ± 70 m diverging scale (98th-ptile auto-scale; the intense Weddell/Ross convection core saturates it). The mixing runs (06N/O/T/V) restructure the Weddell/Ross winter mixing relative to Baseline, whereas melt-pond (06A) barely changes it. Deep open-ocean winter mixing there is FESOM “open-ocean convection” that ventilates warm deep water — a known driver of the Southern-Ocean warm bias (Heuzé et al. 2013; Heuzé 2021) and of TOA imbalance.

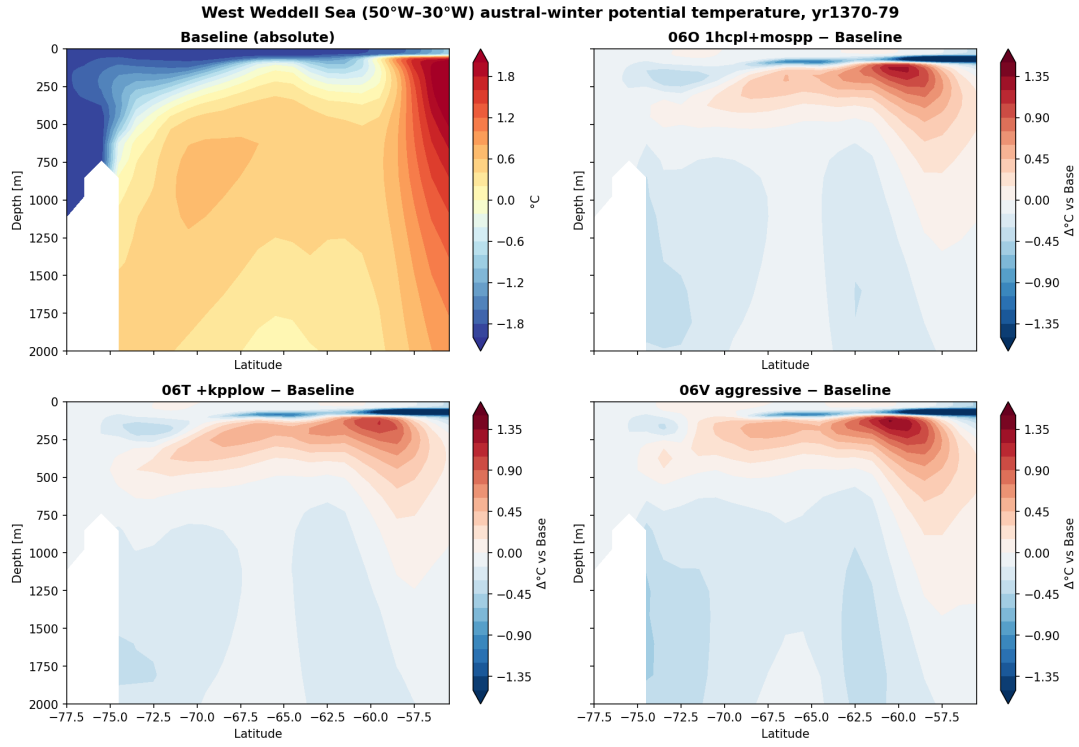


Figure 8: West Weddell Sea (50°W–30°W) September potential-temperature section: Baseline absolute and anomalies. The ventilated mid-column cools in the mixing runs (less convective entrainment of warm deep water to the surface), consistent with suppressed open-ocean convection; note that suppressing convection should leave the *deepest* water trapped/warmer rather than colder, and a near-surface edge warm anomaly is also present — i.e. the signal is a redistribution between layers, not uniform subsurface cooling.

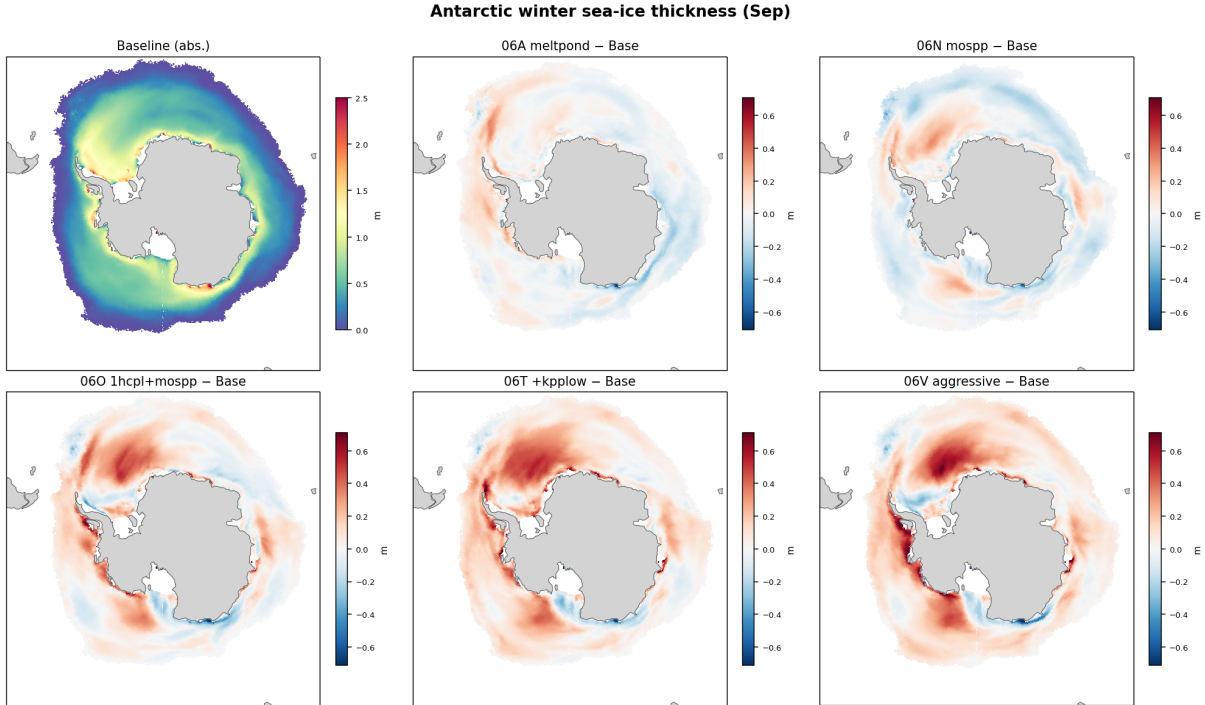


Figure 9: Antarctic September sea-ice *thickness* (Baseline absolute, top-left; others = anomaly vs Baseline). Thickness — not concentration — is the honest growth diagnostic here: winter concentration is already near 1 over most of the pack, so it *saturates* and cannot register added ice, whereas thickness is unbounded. The mixing runs (06O/06T/06V) thicken the pack broadly (positive/red anomaly over the Weddell and Ross sectors), consistent with suppressed open-ocean convection insulating and stratifying the surface; melt-pond (06A) barely changes it. (The concentration version, `sic_winter_antarctic.png`, shows only the ice-edge shifts for the reason above.)

Mechanism. The mospp / weaker-KPP changes reduce Weddell-Sea open-ocean winter convection, cooling the

subsurface and letting sea ice grow — improving `thetao/so/siconc` (consistent with the CMPI breakdown) and removing the Southern-Ocean surface warm bias. Spurious open-ocean deep convection in the Weddell/Ross gyres is the canonical CMIP Antarctic bias (Heuzé et al. 2013; Heuzé 2021, which evaluates AWI-CM in the multi-model set; Mohrmann et al. 2021 for the polynya surface expression), and open-ocean Weddell convection is a real-but-rare mode that models over-express (de Lavergne et al. 2014) — so damping it is physically the right direction. The *proximate radiative* closure is defensible: more winter ice → higher albedo → less absorbed SW → lower imbalance. We are more cautious about the “reduced ocean heat uptake” half of the chain: suppressing convection also *traps* heat at depth, which can *increase* deep ocean-heat-content drift, so the net uptake change is not established here — it should be pinned to an OHC-tendency diagnostic before being asserted. The cost is that the surface cooling propagates globally into the atmosphere (`tas/zg`), the broad cold bias seen in `part8`.

Counter-literature (a compensating-error risk this campaign must own). §5 attributes the SO warm bias and its removal almost entirely to *ocean* convection, but Hyder et al. (2018, *Nat. Commun.*) trace the leading 40–60°S warm-SST bias primarily to *atmospheric* net surface-heat-flux errors dominated by shortwave/cloud biases, and warn that a small *net* flux bias is often achieved by *compensating* component errors. The two coexist at different scales — broad atmospheric-SW warmth vs. localised Weddell/Ross deep-convection anomalies — and our mixing levers act only on the ocean branch. If part of the bias is atmospheric-SW in origin, the ocean-mixing lever is partly compensating for an atmospheric error, which is *exactly* the “cooling leaks beyond where the parameter should act” failure mode flagged for these runs in §4.3 (the region map reddens into the tropics/Niño-3.4). This tension is the report’s own evidence pointing at Hyder; it strengthens, not weakens, the case for treating aggressive ocean-mixing tunings as provisional.

Part II

Counter-tuning the NH cold bias — the structural floor

The cold bias that makes Part I’s aggressive lever expensive is largely **not** the tuning: it is a structural LPJ-GUESS boreal-forest collapse that every run inherits. That makes it a **counter-lever** — fixing it warms the NH high latitudes at essentially unchanged imbalance, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. This part attributes the bias with a clean WITH-vs-WITHOUT-LPJG experiment, resolves its two mechanisms and a residual, and — with a decisive feedback-free spin-up — pins the fix to vegetation competition, not climate.

6 The NH high-latitude cold bias is a structural LPJ-GUESS problem

The runs carry a severe near-surface **cold bias over Siberia and northern Canada** (visible in every part8 map). This chapter documents a working investigation into whether the coupled dynamic vegetation (LPJ-GUESS) is responsible — motivated by the idea that fixing it could be a *warming* lever to offset the imbalance-tuning cold bias, and by prior EC-Earth experience that LPJG drives such boreal biases.

6.1 Symptom: boreal needleleaf replaced by grass

Dominant-PFT maps (tool part26) for the HTESSSEL spin-up (the reference, run with the same atmosphere but *without* atmosphere↔LPJG feedback) vs the coupled runs show boreal needleleaf giving way to C3 grass over Siberia / N. Canada. Tree-dominant fraction north of 50°N: **20.1 % (spin-up) → 16.8 % (baseline) → 11.4 % (06V)** — progressively worse with the more aggressive (colder) runs. *Caveat on the prior*: DGVMs disagree substantially on boreal tree-vs-grass partitioning, and real boreal forest is genuinely open with substantial non-tree ground cover (Jiang et al. 2012), so “grass over Siberia” is not *automatically* a bug. What makes the establishment-failure reading defensible here is the *magnitude*: the coupled state reaches $cvh \approx 0.02\text{--}0.05$ with *no closed canopy anywhere* (FORESTFPC = 0, §9), an outlier well below the $\sim 0.4\text{--}0.7$ closed-taiga range (MODIS VCF), not merely a grass-leaning partition.

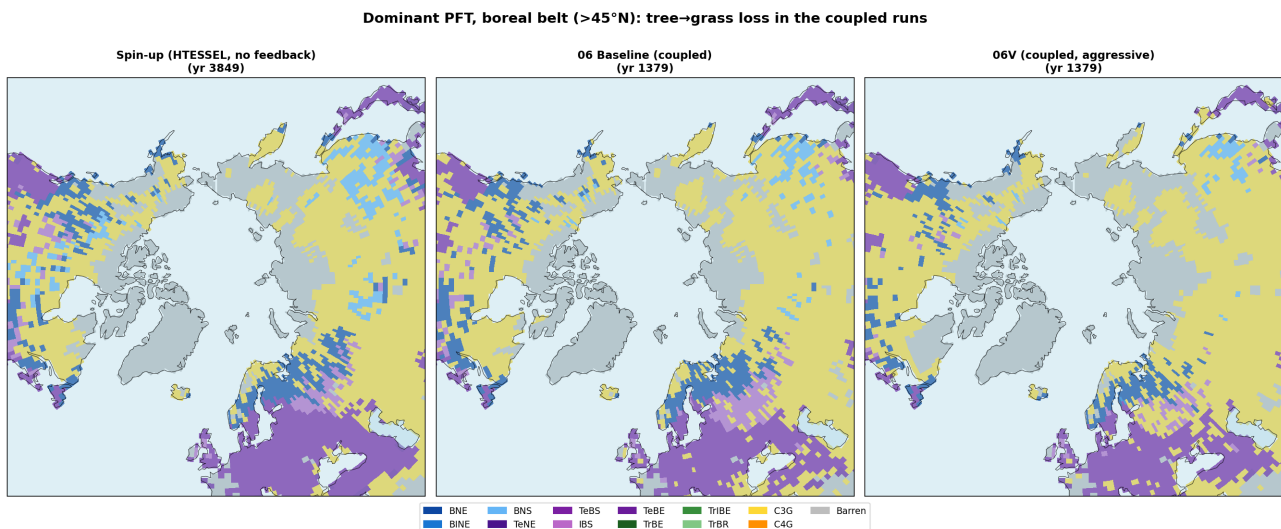


Figure 10: Dominant PFT, boreal belt: HTESSSEL spin-up (reference) vs coupled baseline vs 06V. Needleleaf (blues) thins and C3 grass (yellow) expands across central/eastern Siberia in the coupled runs.

6.2 Is it grass growing, or trees dying? And is it a year-1 jump or a slow drift?

Belt-mean (55–75°N) needleleaf LAI is essentially *conserved* over the baseline run (0.40→0.37, trend ≈ 0), while C3 grass LAI *rises* (0.65→0.83, +0.10/dec): the signal is **grass growing**, with needleleaf loss only regional (E. Siberia) and offset by gains in the milder west. The coupled run is initialised directly from the spin-up’s year-3850 LPJG state, and baseline year-1350 boreal LAI (tree 0.407, grass 0.649) is *identical* to the spin-up end-state (0.401

/ 0.669) — so there is **no wholesale year-1 collapse**; the degraded, grass-heavy state is *inherited from the spin-up*. Within 1350–1379 the boreal 2 m temperature is at its (cold) level from year one and does *not* drift down (trend +0.42 K/dec, noisy) — consistent with a memory-less atmosphere that equilibrates to the surface state within a year rather than a slow vegetation runaway. (These are belt-mean *LAI* diagnostics; as §6.3 shows, the quantity that actually matters is not *LAI-per-tile* but the forest *cover fraction*.)

6.3 The decisive test: a clean WITH-vs-WITHOUT-LPJG pair (CORE3, 30 yr)

The earlier control was confounded by the ocean (core2 vs core3). We removed that confound: a purpose-built **no-LPJG AWI-CM3-develop reference** — *same* CORE3 ocean, *same* TCO95 atmosphere and matched component branches, HTESSEL + prescribed satellite vegetation, a full 30-yr PI-control (model years 1350–1379, `awicm3_noLPJG_CORE3_30y`) — differs from the LPJG Baseline in **LPJG alone**, so the 2 m-T difference is now a clean attribution. Its net TOA imbalance is +1.73 W/m² (**part2**), near the warm/high-imbalance end of Fig. 1 — confirming the LPJG signal is a near-surface *land* effect, orthogonal to the TOA tuning of §2. **Crucially, it beats the LPJG Baseline on both skill metrics: global 2 m-T RMSD vs ERA5 1.514 K (bias −0.76 K, part8) vs 1.717/−0.94, and CMPI 0.738 (part4) vs 0.745** — at higher (warmer) imbalance. Removing LPJG improves the global temperature fit *and* the multivariate skill score *while* the imbalance changes by only +0.2 W/m² (+1.73 vs +1.53) — within the inter-run drift of these unequilibrated 30-yr segments (§7), so the warming lever is real *and costs essentially no imbalance to within that uncertainty* (not, strictly, “zero”). Over the final decade, boreal DJF:

- **Forest cover collapses.** Siberian high-veg cover `cvh`: **0.63 (no LPJG) → 0.02 (LPJG)** (pan-boreal 0.49 → 0.07). LPJG passes essentially no forest to the atmosphere.
- **And it is genuinely colder — confound gone.** With the ocean held fixed, the LPJG run is unambiguously **−3.1 K** colder over Siberia in DJF (−30.4 vs −27.4°C).
- **Snow-masking (spring albedo)** operates as before: forest removal brightens the snow season (MAM +**0.077**). This is the canonical boreal-forest climate mechanism — a masked snowpack lowers albedo and warms high latitudes, and its removal cools most in spring (Bonan, Pollard & Thompson 1992; Betts 2000; Bonan 2008), with tree/canopy cover setting the snow-albedo-feedback magnitude that many models misrepresent (Lorant et al. 2014; Chapin et al. 2005) — so a model that loses its forest gets exactly this feedback wrong. But it is a *sunlight* effect and cannot explain the equally-cold polar night (§6.4).

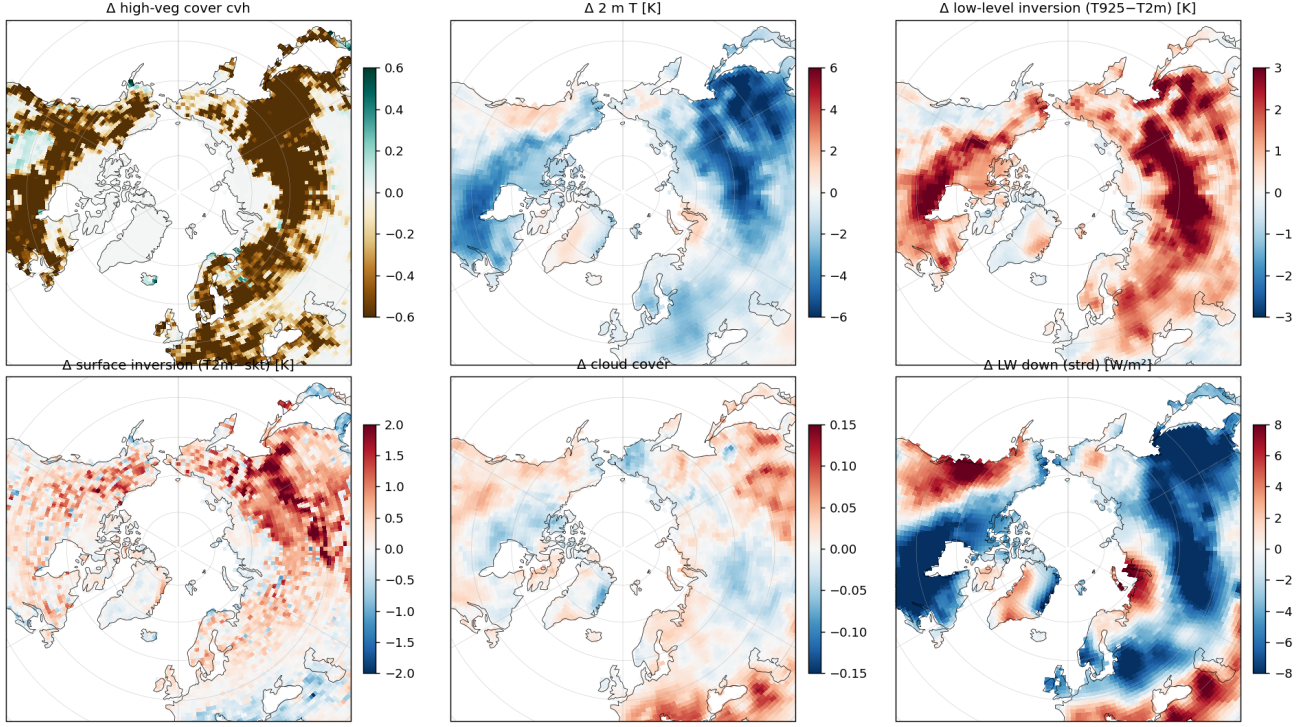


Figure 11: DJF WITH–WITHOUT LPJG, boreal NH (final decade, land-masked). Forest-cover collapse (cvh, top-left, brown) is spatially co-located with the cold anomaly (top-mid), a *stronger* low-level inversion ($T_{925}-T_{2m}$, top-right) and surface decoupling ($T_{2m}-T_{skt}$, bottom-left), at *unchanged* cloud (bottom-mid). LW-down (bottom-right) is modestly reduced over the cold interior.

6.4 Seasonality: which months carry the cold effect

Resolving the same clean CORE3 WITH–WITHOUT pair by *month* (boreal box 55–70°N, Siberia + N. America, final decade) shows the LPJG cold effect is **present in every month except high summer**. The WITH–WITHOUT 2 m-T difference is negative from September through June and only vanishes in July–August ($-0.1/-0.2$ K), peaking in the *shoulder/dark* seasons: **November -3.9 K, April -3.3 K**, with DJF -2.4 K and MAM -2.5 K (annual box-mean -2.0 K). **This is the key seasonal fingerprint:** a pure snow-albedo (sunlit) mechanism would confine the cold to the bright spring, yet the effect is *as large in near-sunless November/December* — which is exactly what motivates the winter roughness→inversion mechanism of §6.5. The near-zero July–August signal also confirms the growing-season atmosphere is unaffected (consistent with JJA gates being cleared, §6.3). Overlaid is the independent **OIFS–AMIP** run (§9.2: observed-SST, no-LPJG, PI 1870–79) — it tracks the *warm* WITHOUT curve (annual -7.6 vs -6.3 °C; DJF -25.6 vs -24.9 ; MAM even milder), confirming that a feedback-free, realistically-forced boreal climate is warm, and that the forcing we drive the decisive offline re-spin with (§9.2) is *not* itself cold-biased.

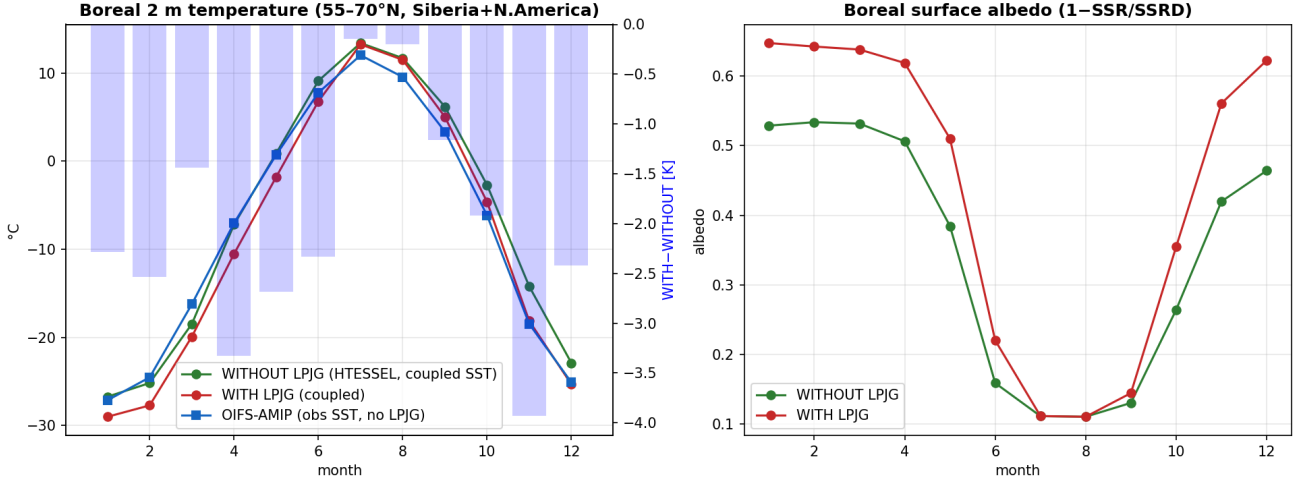


Figure 12: Boreal (55–70°N, Siberia+Canada) monthly climatology. *Left:* 2m-T annual cycle — WITHOUT LPJG (green, clean CORE3 reference), WITH LPJG (red, coupled), and the OIFS-AMIP obs-SST no-LPJG forcing run (blue); blue bars = WITH–WITHOUT [K, right axis], negative (LPJG colder) in every month bar July–August, strongest in Nov/Apr. *Right:* surface albedo (1–SSR/SSRD); LPJG brightens the snow season (spring/autumn), the sunlit part of the bias. (AMIP emits no down-welling SW, so it appears on the temperature panel only. WITH/WITHOUT are model years 1370–79; AMIP is obs-SST 1870–79 — both $\sim$ PI.)

6.5 A second, winter mechanism: roughness $\rightarrow$ inversion

Snow-albedo needs sunlight, yet the cold bias is as strong in DJF as in MAM. The missing forest acts a second way, in the dark. OpenIFS roughness length is cover-weighted by cvh , and needleleaf forest ($z_{0m} = 2.0$ m) is 20–60 $\times$ rougher than the tundra/grass (0.034–0.10 m, `susveg_mod.F90`) that replaces it. A smoother surface couples less efficiently to the atmosphere; in polar night the stable boundary layer (SBL) decouples and the surface radiatively cools, deepening the near-surface inversion. That surface–atmosphere coupling strength in the SBL controls screen temperature and inversion depth is well established (Holtslag et al. 2013; Sandu et al. 2013); we assemble the specific vegetation-roughness $\rightarrow$ SBL-decoupling $\rightarrow$ screen-T chain from the land-surface and SBL literatures rather than a single published result, so it is presented as a mechanism the data then test, not a settled citation. **The data show exactly this fingerprint** (DJF Siberia, LPJG – no-LPJG): a **+2.1 K stronger** low-level inversion ($T_{925} - T_{2m}$), a **+0.7 K stronger** surface inversion ($T_{2m} - T_{skt}$), **unchanged cloud** ($\Delta \approx 0.00$) and **–4.7 W/m² less** down-welling longwave — a decoupled, radiatively-cooling stable surface, not a cloud/radiation artefact. Spatially over boreal land the forest-loss pattern *is* the inversion pattern: area-weighted $r(\Delta cvh, \Delta inv) = -0.60$, $r(\Delta cvh, \Delta sfc\text{-decouple}) = -0.68$, $r(\Delta T_{2m}, \Delta inv) = -0.83$, while the negative control $r(\Delta cvh, \Delta cloud) = -0.05$ is ≈ 0 — a real mechanism, not a shared trend. Regression lever: **restoring cvh by +0.4 warms DJF screen-T by $\sim +1.2$ K** through roughness alone and weakens the inversion by ~ 0.7 K — additive to the spring/albedo warming. This slope-based estimate inherits the scatter of the underlying correlation ($r = 0.63$, so $< 40\%$ of grid-cell variance explained); read it as $\sim +1.2 \pm 0.5$ K, an order-of-magnitude lever rather than a precise prediction. The *direction* and existence of the effect rest on the WITH–WITHOUT experimental design (§6.3), not on this regression.

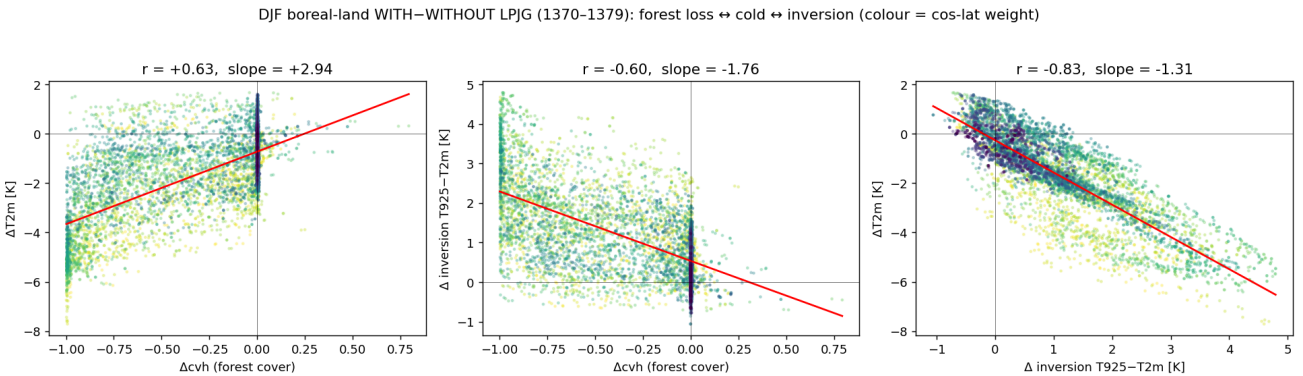


Figure 13: DJF boreal-land, WITH–WITHOUT LPJG (final decade). Forest loss (negative Δcvh) tracks the cold (ΔT_{2m} , left), the stronger inversion (middle) and, transitively, cold $\leftrightarrow$ inversion (right, $r = -0.83$). Colour = $\cos\text{-lat}$ weight.

6.6 How much does the LPJG fix buy? (vs ERA5)

Against ERA5 the current Baseline is -4.5 K too cold over Siberia in DJF. The lever helps but does not close it: a realistic partial fix ($+1\text{ K}$) leaves -3.5 K , and *full* forest restoration (the no-LPJG run itself — the ceiling) reaches -1.4 K . So vegetation/roughness accounts for $\sim 3\text{ K}$, roughly two-thirds of the Siberian DJF cold bias, and boreal-land RMSD falls $4.4 \rightarrow 3.3\text{ K}$ — but a residual $\sim 1.5\text{ K}$ survives *even with perfect forest*.

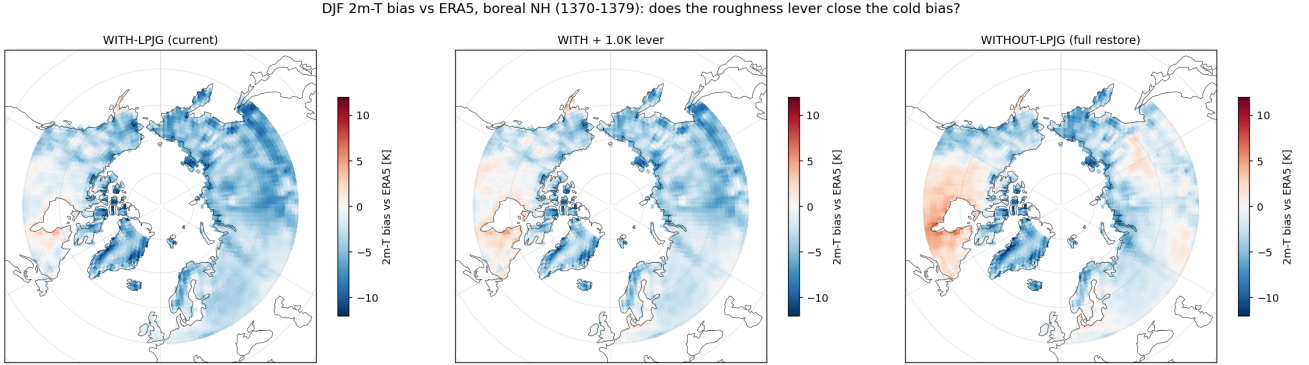


Figure 14: DJF 2m-T bias vs ERA5, boreal NH (final decade): current LPJG Baseline (left, model output), an **illustrative** $+1\text{ K}$ roughness lever (middle — a uniform $+1\text{ K}$ offset applied to the Baseline, *not* a simulation), and full forest restoration = the no-LPJG run (right, model output). The lever panel only visualises the magnitude of a plausible partial fix; the load-bearing comparison is left vs right, which shows a residual cold survives even full forest restoration.

6.7 The residual driver: an over-strong stable-boundary-layer inversion

Diagnosing that residual in the *full-forest* run (vegetation thereby ruled out) against ERA5 (DJFM, boreal land) points to a single process. The model’s low-level inversion is **$+1.4\text{ K}$ too strong** ($T_{925} - T_{2m}$: model $+5.6$ vs ERA5 $+4.2\text{ K}$), and the excess is **entirely surface-trapped** — the 925–1000 hPa inversion bias is $+0.07\text{ K}$ (≈ 0), so all the extra cooling sits in the lowest layer. Spatially this inversion-excess explains $r = +0.76$ of *where* the model is too cold; the Siberian High (MSLP bias $+0.15\text{ hPa}$, $r = 0.27$) and cloud-fraction (bias $+0.07$, $r = 0.00$) are ruled out. A too-strong wintertime polar inversion is a real, observationally-benchmarked model bias (Medeiros et al. 2011; Zhang et al. 2011), which supports reading the “ $+1.4\text{ K}$ too strong” residual as genuine. That surface-trapped signature, with circulation and cloud-fraction eliminated, is also the textbook fingerprint of a **stable boundary layer that under-mixes** — the IFS-family “too-stable” winter cold bias (Sandu et al. 2013). Its lever is the OpenIFS stable-BL turbulent mixing — the “long-tail” stability functions / minimum stable diffusion whose lineage runs Louis (1979) $\rightarrow$ Beljaars & Viterbo (1998) — *orthogonal* to the vegetation fix.

Two counters this diagnosis must own (both align with the report’s own “compensating error” worry). (a) *The mixing fix is a documented compensating error*: enhanced long-tail stable mixing is stronger than LES/observations justify and is retained only because reducing it degrades the large-scale flow — it improves 2 m temperature but deepens cyclones and harms low cloud (Sandu et al. 2013; Holtslag et al. 2013), so Phase 1B must check the storm-track/low-cloud climatology (which CMPI/part8 will not catch cleanly). (b) *The inversion may be a cloud-phase problem*: Pithan et al. (2014) attribute much of the modelled Arctic wintertime cold bias to missing mixed-phase low cloud (a LW, cloud-phase effect) — our null control ruled out cloud *fraction* but not phase, so we scope the residual claim to “given cloud fraction is unbiased” and flag a cloud-phase / LW $\downarrow$ test as a prerequisite. Likewise the $r = +0.76$ residual-to-inversion match is suggestive, not proof that the “fourth lever” is truly orthogonal.

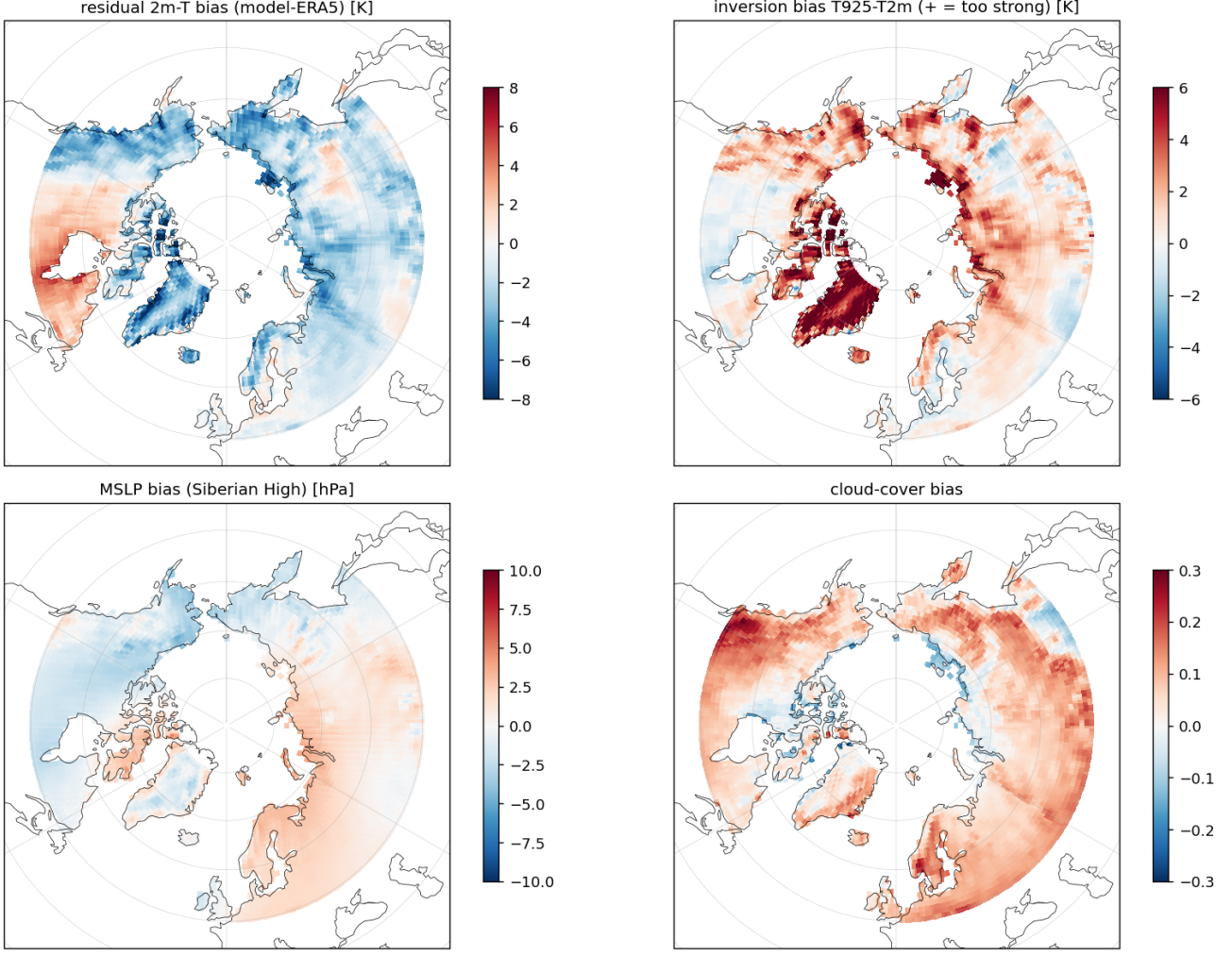


Figure 15: Second-driver search: full-forest (no-LPJG) run DJFM vs ERA5, boreal NH. Residual cold (top-left) matches the inversion-excess (top-right, $r = +0.76$); MSLP (bottom-left) and cloud (bottom-right) biases are diffuse and uncorrelated.

Near-complete attribution. The two mechanisms close the boreal DJF cold bias between them: of the ≈ -4.5 K Siberian bias vs ERA5, the LPJG forest-cover collapse (snow-albedo in spring, roughness in winter) contributes ≈ -3 K, and the over-strong stable-BL inversion the remaining ≈ -1.4 K. Two *orthogonal* levers — the LPJG establishment fix and enhanced stable-BL mixing — therefore target essentially the entire bias. (The veg cooling is also a *stable* offset, not a runaway: over 20 yr the LPJG run carries more/thicker Arctic ice as a bounded mean offset with no amplifying sea-ice drift.)

6.8 Working conclusion & lever

LPJ-GUESS sends essentially no boreal *forest cover* to OpenIFS ($cvh \sim 0.6 \rightarrow 0.02$ over Siberia, §6.3), so the canopy no longer masks snow (spring albedo, §6.3) *nor* roughens the surface (winter inversion, §6.5) — a physically robust driver of the Siberia/N. Canada cold bias, consistent with the EC-Earth family in which LPJ-GUESS is interactively coupled (Döscher et al. 2022) and in which dynamic vegetation-cover / effective-LAI improves high-latitude spring climate through the snow-albedo coupling (Alessandri et al. 2017). Source trace (verified): LPJG computes the high-veg fraction and exports it over OASIS as `GUE_FRAH→cvh` (`framework.cpp::computeDailyLAIandFPCforIFS` namcouple line 120). With the AWI setting `iftreefracca=1` (`global.ins:151`), `cvh` is the summed tree *crown area* $\times$ density (no 0.05 clamp — that constant is dead code; `natural_frac=1` over PI land), so $cvh \approx 0.05$ means boreal tree cover is genuinely $\sim 5\%$. The establishment gates (`gdd5min_est=500`, `twmin_est=5`) are *not* the blocker — they are cleared across the boreal — so this is a **tree-vs-grass competition failure** (grass out-produces trees at canopy closure; settled in §6.9), with the degraded state *inherited from the LPJG spin-up*, not a coupled runaway. Seasonally (§6.4) the LPJG cold effect spans September–June (DJF -2.4 , MAM -2.5 K over the boreal box) and vanishes only in July–August — i.e. it is a cold-season, not a growing-season, effect. The co-equal *sun-*

less late-autumn/winter part — which pure snow-albedo cannot explain — is now resolved (§6.3–6.7): it is the roughness→inversion effect of the lost forest (§6.5) plus a model-intrinsic over-strong stable-BL inversion (§6.7).

Fixes: (1) tune the boreal **tree-vs-grass competition** (`greff_min`, C3-grass competitiveness, cold-tree photosynthesis `pstemp_low`) — *not* the establishment gates, which are correct (§6.9) — and re-do the LPJG spin-up; highest value, cheap to screen offline by checking `cvh` recovers toward the MODIS pattern; (2) revisit `iftreefracca/crown` allometry; (3) a woody stem-area floor on the snow-masking fraction in OpenIFS (`surfbc_ctl_mod.F90:3391 / surfgrad_ctl_mod.F90:650–655`) — but this only bites once (1) raises `cvh`, since the masked-snow tile is weighted by `cvh` itself. Fixes (1)–(3) are the physically-grounded *warming lever* of §8 and buy ~ 3 K (§6.6). **A fourth, orthogonal lever** attacks the residual ~ 1.4 K (§6.7): *enhanced stable-boundary-layer mixing* in OpenIFS (VDF stability functions / minimum stable diffusion), which would relax the over-strong surface inversion. Together (1)–(4) target essentially the entire boreal DJF cold bias.

6.9 Decisive test: competition, not climate

Is the boreal forest deficit a *climate* problem (fix the atmosphere) or an *LPJG-internal* tree-vs-grass competition problem (fix the vegetation)? We answered it directly with a **feedback-free** experiment: a clean OpenIFS–AMIP run (prescribed observed SST, no coupled ocean/LPJG cold feedback; $\sim$ PI 1870–79) generated a realistic daily forcing, which drove a 2000-yr standalone LPJG re-spin on a *dense* 11 538-cell gridlist (build detail in §9.2). Fig. 16 puts that forcing (top) beside the equilibrium vegetation it produced (bottom). **The forcing is *not* growing-season-cold:** over the Siberian box the warmest-month T2m is 14.7°C (100% of cells clear the `twmin_est` = +5 gate) and GDD5 is 730 (76% clear 500) — establishment is climatically permitted essentially everywhere trees should be. **Yet the spin-up under-vegetates and grass wins:** `cvh` = 0.21 (far below the 0.4–0.7 closed-taiga range), `GRASSFPC` = 0.37 > `TREEFPC` = 0.21, and *no closed canopy forms anywhere* (`FORESTFPC` = 0), with grass filling exactly the East-Siberian interior. On a dense gridlist this cannot be dismissed as under-sampling. **Verdict: the deficit is primarily an LPJG-internal competition problem, not a climate one** — so the primary counter-lever is the vegetation model (boreal `greff_min`, C3-grass competitiveness, `pstemp_low`), not another atmospheric retune. The one climate co-limit is the deep-winter cold (-28.3°C box mean, < -40 in NE Siberia) that gates out evergreens on $\sim 40\%$ of cells — but climatically-permitted *larch* (no cold gate) fails there too (BNS cover 0.035), so the winter warming of Phase 1B is only a *secondary* aid.

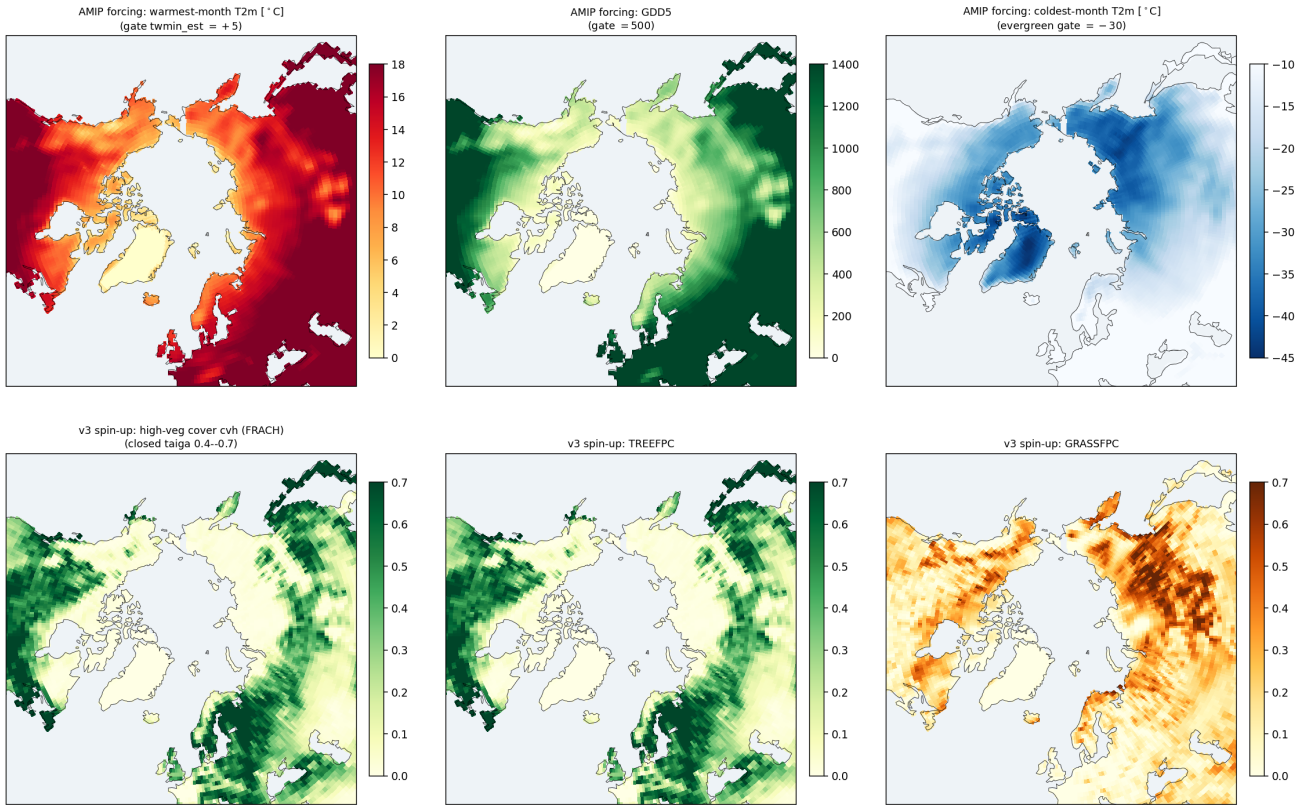


Figure 16: Decisive climate-vs-competition test (NH boreal, equilibrium yr 3900). *Top:* the feedback-free obs-SST ~PI OIFS-AMIP forcing that drove the standalone LPJG re-spin — warmest-month T2m (vs `twmin_est` = +5), annual GDD5 (vs 500), coldest-month T2m (vs evergreen `tcmin_est` = -30). *Bottom:* the equilibrium vegetation it produced — cvh, TREEFPC, GRASSFPC. The growing-season forcing clears the gates across the boreal, yet the spin-up yields $cvh \approx 0.2$ with grass dominating the East-Siberian interior and no closed canopy — grass out-competes trees under an adequate climate. The only climate co-limit is the deep-winter cold (top-right, NE Siberia), but climatically-permitted larch fails there too.

Part III

The path forward: raise the floor, then tune

Parts I-II give the strategy: the imbalance can only be cut by cooling, and cooling is only affordable once the structural NH cold floor is raised. So first fix the floor (vegetation competition + stable-BL mixing), then spend a moderate cooling lever to reach surface balance — now without the RMSE blow-up. First the caveats that condition this, then the recommendation and a phased plan.

7 Caveats

1. **06D “HRlike” has no HR-specific namelist change** — only the common pond shift (identical to 06C); its metrics confirm this. If a resolution/timestep/viscosity change was intended, it is not in the namelists.
2. **Filename convention.** These runs write `atm_remapped_1m_<var>_YYYY-YYYY.nc` (no `_1m_infix`). I added backward-compatible fallbacks to `part2`, `part8` and the CMPI preprocessing; existing configs are unaffected.
3. **Accumulation period** for these XIOS runs is **3600 s** (not the 21600 s of the old HR config); configs use `accumulation_period=3600` (verified: $tsr \rightarrow \approx 240 \text{ W/m}^2$).
4. None of the runs is at equilibrium within 30 yr — numbers are a ranking, not final values.

8 Recommendation & synthesis

Reading global CMPI/RMSD as a guide rather than a gate (the PI-vs-PD offset is expected), the picture is:

- The **melt-pond / albedo family (06A–06D, etc.)** are *clean*: they improve sea ice where they should and leave everything else essentially untouched. Their imbalance gain is modest (5–20%) but it comes with no unexpected collateral — safe building blocks.
- The **ocean-mixing / coupling family (06O/T)** delivers the only large imbalance reductions (30–38%) *and* genuinely improves the ocean interior and sea ice — but the cooling leaks into the atmosphere globally (incl. tropics), which is degradation in places the parameter should not reach. That makes them promising but not yet “free.”
- **06V** pushes too far: the ENTSTPC3 convection change degrades **tas/zg** everywhere — not recommended as-is.

Best-balanced single steps: **06P** (–29%, well-localised) and **06H** (–20%).

Synthesis — and where the warming lever comes from. The two arcs join here. The imbalance can only be cut substantially by cooling levers (ocean mixing/coupling), so every combo tried so far overshoots into cold bias. But §6 shows the shared Siberia/N. Canada cold lobe is *not* a tuning artefact, and a purpose-built clean no-LPJG CORE3 reference (§6.3) now *quantifies* it: of the ≈ -4.5 K Siberian DJF bias vs ERA5, ≈ -3 K is the LPJG forest-cover collapse (via snow-albedo in spring *and* surface roughness in winter, §6.5) and ≈ -1.4 K is a model-intrinsic over-strong stable-BL inversion (§6.7). The clean run is *better* than the Baseline globally (RMSD 1.514 vs 1.717 K) at *near-unchanged* imbalance ($+0.2$ W/m², within drift) — indicating both are *warming levers largely orthogonal* to the TOA tuning. That reframes the path forward:

1. Fix the LPJG boreal forest-cover bug (establishment limits `gdd5min_est/twmin_est` and/or zero-LAI stem snow-masking — see the companion note) and re-do the spin-up; this warms the NH high latitudes in *every* run, lowering the shared RMSD floor by ≈ 3 K in the worst region.
2. Attack the residual with *enhanced stable-BL mixing* in OpenIFS (§6.7) — an independent knob that relaxes the surface-trapped winter inversion.
3. On that warmer baseline, apply a *moderate* mixing/coupling setting (06O-like, not 06V) for the imbalance, plus the clean melt-pond changes — the cooling it adds is then offset rather than additive.
4. Re-judge on the *pattern* of CMPI change (§6.3 method), not the global value, since the PI-vs-PD offset is expected.

In short: don’t chase the imbalance with ever-colder atmospheres; remove the structural cold bias first, then the moderate cooling needed for the imbalance becomes affordable.

9 Work plan: a systematic path to a warm-floor, surface-balanced PI control

Two revised targets. (i) net *surface* energy balance $|F_{\text{sfc}}| \lesssim 0.3$ W/m² — *not* TOA. Surface and TOA net flux track to within ≈ 0.1 W/m² in every run (§4.1); they are one problem, so a surface-zero target is equivalent to TOA ≈ -0.1 W/m², marginally stricter than TOA = 0. (ii) global 2m-T bias vs ERA5 ≈ -1.3 °C — the physically-expected PI-minus-present-day offset, *not* zero: a run that reaches -1.3 °C is *correct*, not cold-biased.

Ordering principle. The imbalance can only be cut by cooling, and every cooling lever over-cools if applied to today’s cold-biased base. So we *first* raise the warm floor with two structural fixes that are *orthogonal to the TOA balance* (they change the land surface and BL mixing, not the radiation budget — §6.3 showed the no-LPJG floor sits at *near-unchanged* imbalance, $+0.2$ W/m² within drift), *then* spend a *moderate* cooling lever to reach surface balance, now affordable because the floor is ~ 3 – 4 K warmer where it was worst. Phases 1A/1B are independent (parallel); Phase 2 depends on both; Phase 3 gates.

9.1 Phase 1A — Restore boreal tree cover (LPJG establishment)

Literature & observational basis — what realistic trees require (so we don’t green the tundra). The observed arctic/boreal treeline sits near the July 10 °C isotherm (Köppen), refined by Körner & Paulsen (2004) to a growing-season mean of ~ 6 – 7 °C over ~ 90 days, with the northern boreal limit at $\lesssim 30$ days > 10 °C. **Crucially, LPJG’s current gates are already the standard, treeline-calibrated values** (Smith et al. 2001; Sitch

et al. 2003 — byte-identical to the canonical `global.ins`): BNE/BINE `twmin_est`= 5 (already *below* the 10°C isotherm) and `gdd5min_est`= 500 sit essentially *at* the observed treeline — so loosening them risks trees north of the real treeline. Two further realism facts constrain the fix: (i) NE Siberia’s forest is ~80 % *larch* (deciduous BNS), because evergreen conifers cannot survive its extreme winters (Kharuk et al.; Herzsuh 2020, *Glob. Ecol. Biogeogr.* 29, 198–206, the deciduous-vs-evergreen distribution paper) — LPJG already encodes this (BNE/BINE `tcmin_surv`= –31 excludes them; BNS has *no* cold gate and `gdd5min_est`= 350), so the realistic tree there is larch, and it is *already permitted*. (ii) Closed taiga tree cover is ~0.4–0.7 (MODIS VCF), falling below 0.2 at the taiga–tundra ecotone — the spatial shape `cvh` must reproduce.

Larch caveat. The realistic NE-Siberian tree is *deciduous* larch, whose leafless winter canopy masks little snow — so the spring *albedo* mechanism is genuinely weaker over the larch zone (though it retains full force over the evergreen west Eurasia / N. America). This only re-weights the two mechanisms, it does not soften the conclusion: the *roughness* mechanism survives (even leafless, larch is $\gg$ rougher than grass/tundra), and the seasonality fingerprint (§6.4) shows the LPJG cold effect is co-equal in the *sunless* Nov–Jan months — i.e. the dark-season roughness/inversion part is what carries the larch zone.

The coupled winter cold is a self-inflicted trap, not the trigger. At the cells where prescribed veg has forest but LPJG does not (1785 boreal-land cells), the growing-season gates are already cleared in every run ($\text{JJA} \geq 12^\circ\text{C}$, $\text{GDD5} > 500$); only the winter `tcmin_est`= –30 evergreen gate binds. But against the *forested* WITHOUT-LPJG climate it excludes evergreens on just 23% of cells ($\approx$ ERA5’s 22%), vs 43% in the degraded WITH-LPJG run — the missing forest’s own -3.3°C winter cooling doubles the exclusion. So it is a **bistable trap** seeded by the under-forested spin-up, not a climate trigger.

The decisive test settles it (§6.9): competition, not climate — so Phase 1A’s lever is the **competition parameters**, not the establishment gates: boreal `greff_min` (0.03 BNE / 0.09 BINE,BNS), C3-grass competitiveness, and cold-tree photosynthesis `pstemp_low`= 10°C. Robustly, trees *do* establish (nonzero density) and fire is never involved (burnt fraction = 0 despite BLAZE) — the failure is canopy-closure/productivity competition, not a gate or a fire problem. Phase-1B winter warming (§9.3) is a *secondary* aid, only on the coldest evergreen-gated cells.

Why not just relax the gates? The summer gates are not binding (above), so loosening them cannot help; and the *one* binding gate — the winter `tcmin_est` — correctly excludes evergreens from the extreme-cold larch zone, so relaxing it would plant *unrealistic* evergreens where even ERA5 says larch. The fix is therefore the winter *climate*, not the limits. (Keep `ifreefracca` 1, `global.ins`:150: `cvh` is the summed tree *crown area*, so it tracks establishment success directly.)

Screen & deploy. Tune the competition knobs on the cheap standalone LPJG spin-up (the OIFS–AMIP forcing of §6.9; minutes–hours of wall-time), targeting the MODIS-VCF pattern (~ 0.4 – 0.7 closed taiga, < 0.2 at the ecotone, larch winning the cold NE) and a closed canopy (`FORESTFPC` > 0, tree NPP > grass NPP) — *not* merely a higher global mean. On a passing screen, re-do the coupled LPJG spin-up, regenerate the restart, and carry it into Phase 2.

9.2 Execution: the clean AMIP–noLPJG forcing generator

The decisive test (§6.9) needs a *feedback-free* climate to drive it — reusing coupled-LPJG output would carry the very cold bias we are testing. We therefore built a dedicated **OpenIFS–AMIP** generator at TCO95 (`oifsamip-cy48`, git-aligned to `awiesm3-develop`): AMIP prescribes observed SST/sea-ice, so the atmosphere has no coupled ocean/LPJG feedback. Building and running it required fixing three esm-tools bugs (a FESOM-less crash, missing XIOS/coupling compile flags, a missing WAM date-preprocessing recipe — all committed & pushed; detail in App. B). A 10-yr $\sim$ PI decade (1870–79 observed SST, GHG fixed 1850) produced the daily forcing. *One methodological catch worth recording:* the first re-spin came out barren everywhere because the XIOS field-def de-accumulated fluxes over the 6 h `freq_op` cadence instead of the true `NFRHIS`×`TSTEP`= 1 h period, making `rsns/rlns/pr` exactly 6× too low (net SW 28 vs 166 W/m²) — fixed by a validated ×6 patch and, at root, by templating the divisor from esm-tools. **Expected payoff** of the whole Phase 1A (§6.6): $\approx -3\text{ K}$ off the Siberian DJF bias, $\approx -0.2\text{ K}$ global RMSD, at near-unchanged imbalance.

9.3 Phase 1B — Enhance stable-boundary-layer mixing (OpenIFS)

Change. The residual $\approx 1.4\text{ K}$ surface-trapped inversion (§6.7) is relaxed by strengthening stable-BL turbulent mixing in the EDMF K-diffusion scheme via the “long-tail” (LTG) stability functions — lengthening the tail keeps mixing non-zero at large Richardson number and weakens the over-strong surface inversion. There is no namelist for the stable branch, so this is ~ 2 single-line OpenIFS source edits (minimum stable mixing length `ZLMIN` and the LTG coefficient `ZCB`) plus recompile; the exact file:line targets and secondary companions are in App. B. *Bear the documented side-effects in mind* (§6.7): long-tail mixing beyond the observational range can deepen cyclones

and degrade low cloud, so verify the storm-track/low-cloud climatology, and pre-test cloud phase / LW↓ (Pithan et al. 2014) since the residual may be partly a cloud-microphysics rather than turbulence error.

Test. Change *one* knob first (ZLMIN) to keep attribution clean; scan 2–3 values with short (1–5 yr) coupled sensitivity runs of the no-LPJG reference. **Gate:** the DJFM low-level inversion bias (`djf_second_driver.py`) moves from +1.4 K toward ERA5’s +4.2 K level and the residual boreal cold shrinks, *without* eroding legitimate inversions or warming the tropics (judge globally via `part8/CMPI` pattern). **Risk:** over-mixing warms too broadly / degrades low cloud — mitigate with small steps and global CMPI-pattern judging.

9.4 Phase 2 — Moderate imbalance tuning on the warm floor

With the floor raised (1A+1B) and global skill already better than Baseline (RMSD 1.514, CMPI 0.738; §6.3), spend a *moderate* cooling lever to bring the *surface* budget to zero. Build the combined config in layers, cleanest-first:

layer	lever(s)	ΔTOA [W/m ²]	raw bias [K]	rationale
base	melt-pond (06A–06C; opt. 06G)	−0.08 to −0.26	−1.0 to −1.3	collateral-free
localised	06P (ENTSTPC3)	−0.44 (−29%)	−1.36	best efficiency ~ 1.05 W/m ² /K
moderate	06O (1hcp1+mospp)	−0.55 (−36%)	−1.77	improves ocean/ice; <i>not</i> 06V
(spare)	+kpplow = 06T	−0.58 (−38%)	−1.79	if more cut needed

Why this stops short of 06V. 06V (entstpc3+kpplow all stacked) gives the biggest cut (−57%) but overshoots to −2.1 K and degrades `tas/zg/rlut` in *every* region including the tropics. The base+06P+06O stack cuts ~ 1.0 – 1.1 W/m² of surface imbalance (reference surface +1.52 $\rightarrow \sim 0.4$ – 0.5 ; iterate the coupling strength or add kpplow to close the last ~ 0.4). **Cold-bias accounting:** the ~ -1.7 K raw cooling of the coupling lever, applied on a floor lifted ~ 3 – 4 K where the bias was worst (and ~ 0.2 – 0.4 K globally), lands the global mean near the target -1.3°C rather than the -1.8 to -2.1°C the same lever produces today.

9.5 Phase 3 — Verification protocol and decision gates

Run the combined config 30 yr (evaluate the last decade). Judge with the `reval` toolchain *plus* the DJF driver diagnostics:

target	metric (tool)	accept
surface balance	<code>part2</code> “Surface total”	$ F_{\text{sfc}} \lesssim 0.3$ W/m ² (TOA ≈ -0.1)
cold bias	<code>part8</code> 2 m bias vs ERA5	$-1.3 \pm 0.3^\circ\text{C}$; RMSD $\leq$ ref. 1.51
multivariate skill	<code>part4</code> CMPI	$\leq$ Baseline 0.745; <i>pattern</i> , not just global
no new regionals	CMPI by region (esp. tropics)	no family-typical global cooling leak
driver check	<code>djf_second_driver.py</code> / ...roughness...	inversion bias $\rightarrow$ ERA5; cvh restored

Iteration & critical path. Each axis adjusts independently (the floor fixes are orthogonal to the imbalance lever): too cold $\rightarrow$ back off coupling; imbalance too high $\rightarrow$ add kpplow; inversion too strong $\rightarrow$ raise ZLMIN. Phases 1A and 1B run in parallel now (both cheap/offline); 1A’s coupled re-spin is the long pole, and the Phase-2 run starts once the re-spun restart and the chosen stable-BL build are both in hand.

All scripts, data, figures and this report live in `/work/bb1469/a270092/eval/`: `data/MASTER_metrics.csv`, `data/cmipi_by_variable.csv`, `data/cmipi_by_region.csv`, `configs/`, `plots/`, `run_figures/<run>/`, `notebooks/`. Tool: `release_evaluation_tool2` (reval env); per-run diags via `part2_rad_balance.py` / `part8_t2m_vs_era5.py` / `part4_cmipi.py`.

A Parameter catalogue (distinguishing change per run)

Run	Change (file/namelist)	Baseline → New	Process
06A	albpnd (namelist.ice)	0.2 → 0.28	melt-pond albedo
06B	+ rfracmax	1.0 → 0.75	pond water retention
06C	+ pndaspect	0.8 → 1.3	pond depth:area (=common pond)
06D <i>HRlike</i>	(only common pond shift)	–	no other diff found
06G	albpnd/rfracmax/pndaspect	0.35/0.6/1.6	stronger meltpond
06L	albpnd/rfracmax/pndaspect	0.4/0.55/1.8	extra-strong meltpond
06H	strong pond + RVICE (&NAMCLDP)	0.16 → 0.18	meltpond + cloud-ice fall speed
06Q	albw/albocn	0.1 → 0.045	open-water/ocean albedo
06R	albw/albocn	0.1 → 0.075	open-water/ocean albedo
06S	whichevp (&ice_dyn)	1 → 0	sea-ice rheology mEVP→EVP
06E	RVICE+GGAUSSB (&NAMGWMS)	0.18; -0.5 → -0.6	cloud-ice + gravity-wave
06F	LRDALB (NALBEDOSCHEME=3)	F → T	diagnosed surface albedo
06P	ENTSTPC3 (&NAMCUMF)	absent → 1	deep-convection entrainment
06N	use_momix (namelist.tra)	F → T	near-surface ocean mixing (mospp)
06M	OASIS coupling period + LAG	7200 → 3600 s	atmos↔ocean 2h→1h (1hcpl)
06O	1hcpl + mospp	–	coupling + mixing
06O4	O + momix_kv	0.01 → 0.012	+ larger M-O mixing
06O5	O + momix_kv	0.01 → 0.02	+ even larger mixing
06T	O + KPP-low (kpp_av0/kv0 0.005 → 0.003, bckg ↓ 10×)	–	+ weaker KPP mixing
06U	T + open-water albedo 0.075	–	combo
06V	T + ENTSTPC3=1	–	combo (most aggressive)

B Implementation & lab-notebook detail

Operational detail moved here to keep the work-plan science legible. Retained for reproducibility.

A.1 esm-tools fixes for the AMIP–noLPJG generator (§9.2). Model tree oifsamip-cy48 git-aligned to awiesm3-develop (oifs local_combined_fixes @ 1303a90, oasis afac6a3, xios 2.5.3) + bundle/field patches. Three bugs fixed, committed & pushed (commit 9921d95b, branch development_interactiv_mesh_awiesm3, and a follow-up for the WAM recipe):

- xios.yaml dereferenced the fesom namespace via choose_fesom.* and crashed *any* FESOM-less setup — now gated on "\$(('fesom' in config))".
- oifsamip.yaml never overrode the OIFS comp_command, so the binary compiled *without* XIOS or coupling despite LXIOS:TRUE — now adds -with-cplng2 -with-oifs-xios -build-type=release (like awicm3/awiesm3).
- WAM aborted (CDATEA≠CDATEF) at any start year ≠1990 because the OIFS preprocess step (change_icm_date.sh, which re-dates ICMGG/ICMSH/wave init files) never ran: oifsamip.yaml lacked prepcompute_recipe: \${oifs.preproc} and fell back to the default recipe with no preprocess step. Added (matches awicm3.yaml).

Runscript oifsamip-cy48-levante-TC095L91_lpjgforcing.yaml overrides XIOS output to file_def_lpjg_spinup + field_def_lpjg_safe, emitting daily tas,tasmin,tasmax,pr,rsns,rlns,sfcWind + tdps,ps,psl,clt,prsn,evspsbl,pesm_master gotcha: the single-component target is recomp-<setup>/oifs-48r1 (the *component* name); the alias /oifs silently falls back to the setup and doubles the path (oifsamip-cy48/oifsamip-cy48).

A.2 Flux-accumulation fix (§9.2), at root. The de-accumulation divisor in field_def_lpjg_safe.xml was the hard-coded /21600 (6 h). It is now templated: field_def_lpjg_safe.xml.j2 uses /{{oifs.flux_accumulation_period}} with oifs.flux_accumulation_period defaulting to \${oifs.time_step} (= NFRHIS×TSTEP; with NFRHIS=1 this is the timestep). A validated ×6 scalar patch (AMIP_noLPJG_1d_1870-1879_TC095_PI_fluxfix.nc) corrected the existing forcing without a rerun.

A.3 Phase-1B OpenIFS source edits (§9.3). Paths under oifs-48r1/ifs-source/arpifs/phys_ec/:

knob	file:line	current	proposed	effect
ZLMIN	vdfexcu.F90:201	30 m	40–50 m	min stable mixing length
ZCB	vdfexcu.F90:190	5.0	3.0–4.0	LTG <i>b</i> : lower ⇒ longer tail ⇒ more <i>K</i>

Secondary companions if needed: the 0.1 / 300 m stable mixing-length limits (vdfexcu.F90:374,376) and the surface-layer constants RCHETA/RCHETB/RCHBHD (surf/module/suvexcs_mod.F90, mirror in suvdfs.F90).

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Citations added in response to internal review (rev. 2026-07-06); bibliographic details for Beljaars & Viterbo 1998, Jiang et al. 2012, Medeiros et al. 2011 and Zhang et al. 2011 to be page-verified.